

EVERYTHING IN ITS PLACE: BENCHMARKING SPATIAL INTELLIGENCE OF TEXT-TO-IMAGE MODELS

Wang et.al,

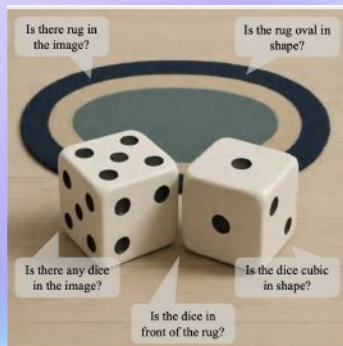
Presented by: Angie Diaz & Nimra Nayyar

01

TEXT TO IMAGE
MODELS CURRENTLY

CURRENT CAPABILITIES OF T2I GENERATION

- State of the art models T2I models exhibit strong computation capabilities in generating specified objects and binding attributes. They have evolved from early GANs to diffusion and multimodal architectures, often supported by LLM text encoders.
- Current benchmarks focus on the “what” of a scene, verifying presence of objects, basic attributes, and simple compositional alignments.



Prompt Type: Long & Sparse Info.

In the scene before us, where simplicity meets subtle charm, the cubic dice, with their sharp edges and intriguing numerical faces, are strategically positioned in front of the elegant oval rug, whose soft contours and inviting textures contrast beautifully with the stark geometry of the dice, creating a harmonious balance that captivates the observer's gaze.

Question Type:

Yes-or-No selections (numbers: five or less)



Prompt Type: Long & Dense Info.

A double exposure portrait features a woman with dark hair in profile, facing left. Her silhouette is filled with a landscape: on the right, the sun half behind of a mountain. On the left, two bare trees stand in the center of a river, the right one taller. Five birds fly toward the mountain in a “V” shape, and bright sunlight casts sharp shadows of the trees.

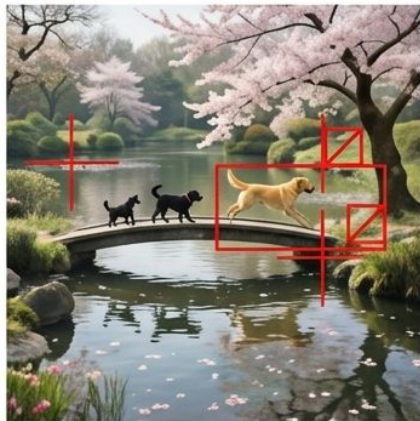
Question Type:

Omni-dims multi-choice questions (numbers: all of 10)

CURRENT CAPABILITIES OF T2I GENERATION

- However, when the task shifts from “what” is a scene to “where”, “how” and “why”, fundamental cognitive blueprints fail.
- Models lack spatial intelligence while benchmarks are limited by short or information-sparse prompts that rely on simple yes/no questions or object detection.

Perception & Layout



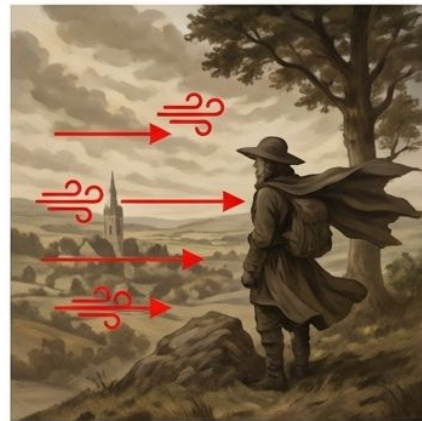
Two small black dogs and a large yellow dog leap across the bridge to the right, one after another.

Spatial Reasoning



The bride's bouquet has ten white roses, five times than the red roses in groom's lapel.

Spatial Interaction



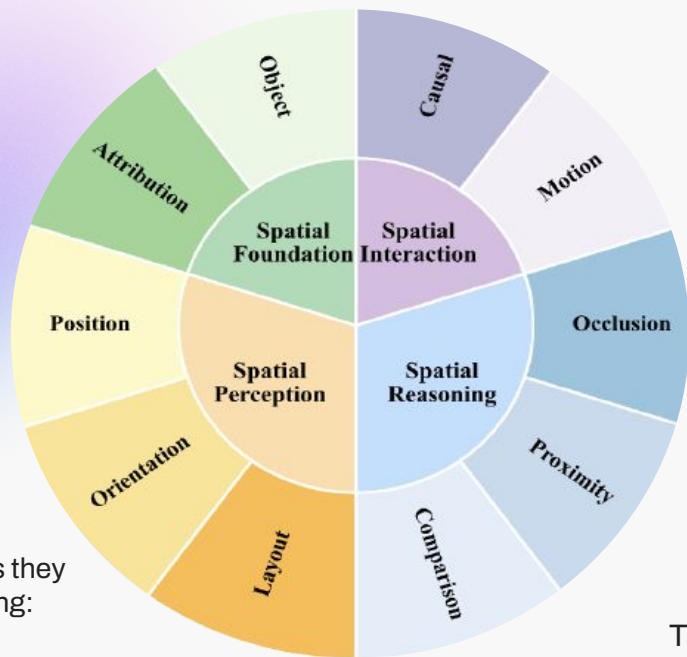
The wind is blowing, causing the traveler's cloak to billow behind them.

02

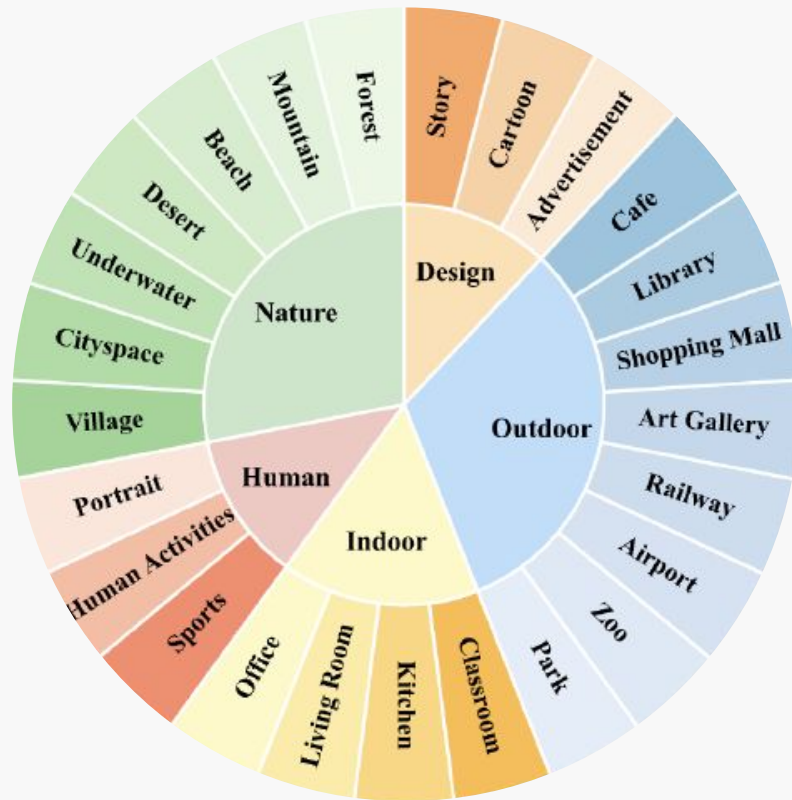
THE EXPERIMENT

SpatialGenEval Benchmark

A new systematic benchmark featuring 1,230 long, information-dense prompts across 25 real world scenes.



First, the main categories they define for spatial reasoning:



The different world scenes for image generation

SpatialGenEval Benchmark

Layer 1 — Spatial Foundation

- Object Identity: Is this a chair or a stool?
- Object Attributes: color, size, material, shape

Prerequisite for all higher layers. Models generally handle this reasonably well.

Layer 2 — Spatial Perception

- Position: Where is the object in the scene?
- Orientation: Which direction is it facing?
- Layout: How are multiple objects arranged?



SpatialGenEval Benchmark

Layer 3 — Spatial Reasoning

Comparison: Which is taller / closer / larger?

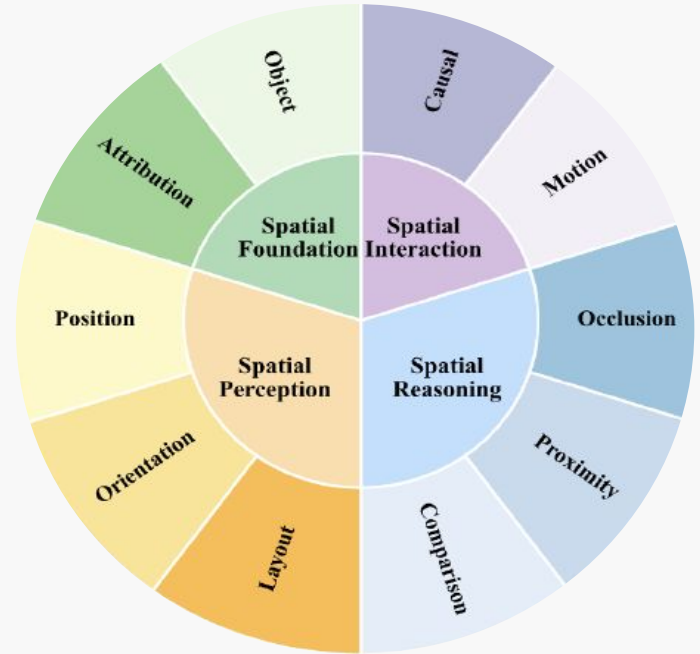
Proximity: How far apart are A and B?

Occlusion: What is hidden behind what?

Layer 4 — Spatial Interaction

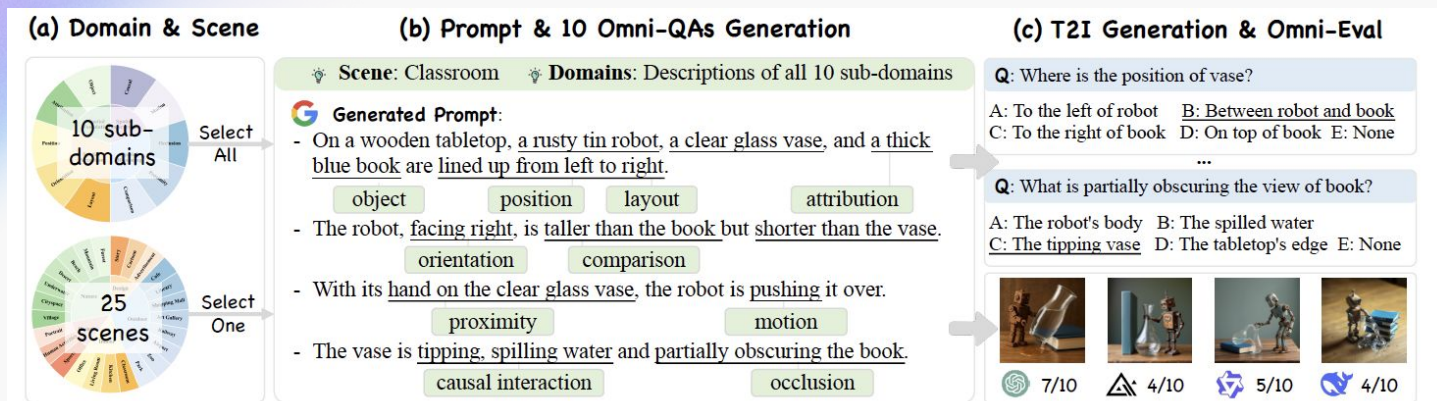
Motion: What is moving, and how?

Causality: Why is something in a state?



SpatialGenEval: Flow

- **Prompt Generation:** Gemini 2.5 Pro is used to synthesize prompts that integrate all 10 spatial constraints into a single, fluent description
- **Image Generation:** 23 different models (diffusion, autoregressive, and unified) generate images from these dense prompts
- **Omni-dimensional Evaluation:** Each prompt is paired with 10 multiple-choice questions (one per sub-domain)
- **MLLM as Judge:** An open-source MLLM (Qwen2.5-VL-72B) acts as the evaluator, selecting the best match from five options (A-E, where "E: None" accounts for generation failures)
 - A 5-round voting mechanism ensures the stability of the results



S1: Object Category



Prompt:

In a library, **two children** sit back-to-back on a colorful **rug** in the foreground. <...>, holds a large picture **book**, <...>. A **toy car** rolls from the boy <...>, its path partially blocked by a plush **teddy bear**.

Q: Besides the two children, what other object categories are mentioned in the scene?

A: Rug, books, toy car, and teddy bear

C: Backpack, pencil, crayon, and paper

B: Window, curtain, plant, and clock

D: Shelf, lamp, table, and chair E: None

S2: Object Attribution



Prompt:

Cartoon: <...>. On the left, a green snail is stretching to cross the red finish line tape in the center. The **brown, cracked-shell** snail on the **right** is far behind, both facing the finish line. <...>.

Q: What are the described attributes of the snail on the right?

A: Green and smooth-shelled

C: Grey and slimy

B: Brown and cracked-shelled

D: Red and small

E: None

S3: Spatial Position



Prompt:

In a park's BBQ area, <...>. **To the left**, a family of four sits at a picnic table. <...>

Q: From a third-person view, where is the family at the picnic table located?

A: On the right

B: In the front

C: To the left

D: In the center

E: None

S4: Spatial Orientation



Prompt:

In a sunlit living room, <...>. On a central jute rug, a toddler crawls **towards three colorful blocks** <...>.

Q: Based on the toddler's movement, what is the toddler facing towards?

A: The ficus tree

B: Away from the rubber duck

C: The colorful blocks

D: The window

E: None

S5: Spatial Layout



Prompt:

Under a giant oak tree, an old storyteller sits on a stool in the center, facing five children sitting **in a semi-circle** <...>.

Q: How are the children arranged around the storyteller?

A: In a straight line

B: In a square formation

C: In a semi-circle

D: In a full circle

E: None

S9: Spatial Motion Interaction



Prompt:

<...>, a female player in a yellow bikini is in the center, **jumping to spike a white volleyball** over the net on the left. <...>.

Q: What is the primary motion described between the player and the volleyball?

A: The player is serving the volleyball B: The player is bumping the volleyball

C: The player is setting the volleyball D: The player is jumping to spike the volleyball

S10: Spatial Causal Interaction



Prompt:

Cartoon: Under a spotlight, a large opera singer in a tuxedo stands center stage. <...>. The **sound wave hits the glass, causing it to shatter**.

Q: What is the direct cause of the glass shattering?

A: The singer touching it

B: The heat from the spotlight

Benchmarks	Prompt Length	Omni-Eval	QA Type	Object	Attribute	Position	Orientation	Layout	Comparison	Proximity	Occlusion	Motion	Causal
T2I-CompBench	S		Yes/No										
GenEval	S		Detect										
DPG-Bench	L		Score										
Wise	S		Score										
TIIF-Bench	L		Yes/No										
OneIG-Bench	L,S		Yes/No										
SpatialGenEval	L		Multi-Choice										

[illegible]

03

FINDINGS

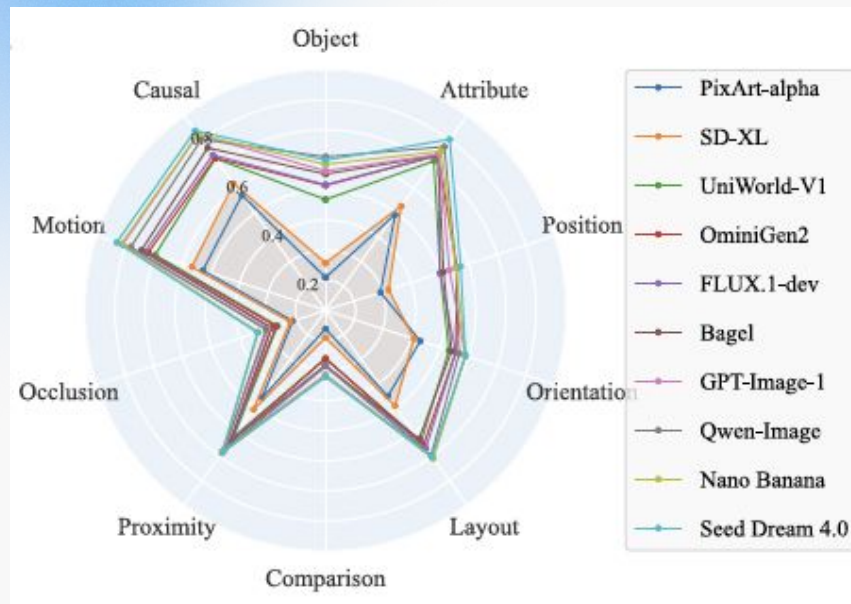
Main Findings

The global ceiling is ~60%

Even SOTA models answer 4 in 10 spatial questions wrong. No model approaches human-level spatial understanding.

- **High scores:** Object identity (~70-80%), attributes (~70%), basic position (~55-65%)
- **Low scores:** Comparison, proximity, occlusion, causality all fall below 30% -- near random chance

Key insight: Models label scenes but do not reason about them -- they know objects exist but not how they relate in 3D space or cause effects



Main Findings

- **Open-source vs. closed-source:** The gap is narrowing. Best open-source model Qwen-Image (60.6%) nearly matches the top closed-source model Seed Dream 4.0 (62.7%)
- **The power of text encoders:** Models with LLM-based encoders (T5, Qwen) consistently outperform CLIP-based ones -- they better parse dense, complex spatial prompts
- **Unified models beat pure diffusion:** Models combining generation + understanding (GPT-Image-1, OmniGen2) consistently outperform diffusion-only models (SD-XL, FLUX)
- **Scaling alone does not solve it:** Even the largest closed-source models cap near 62%. More parameters do not fix higher-order spatial reasoning

The Global Ceiling: ~62%

SOTA models are barely clearing a basic passing threshold for true spatial intelligence.

Rank	Model	Type	Overall Spatial Score
1	Seed Dream 4.0	Closed	62.1%
2	Qwen-Image 20B	Open	60.6%
3	GPT-Image-1	Closed	59.2%
4	Bagel 7B	Open	57.0%
5	FLUX.1-dev	Open	56.5%
6	SD-3.5-L	Open	54.0%
7	DALL-E-3	Closed	51.8%

Error Type Breakdown

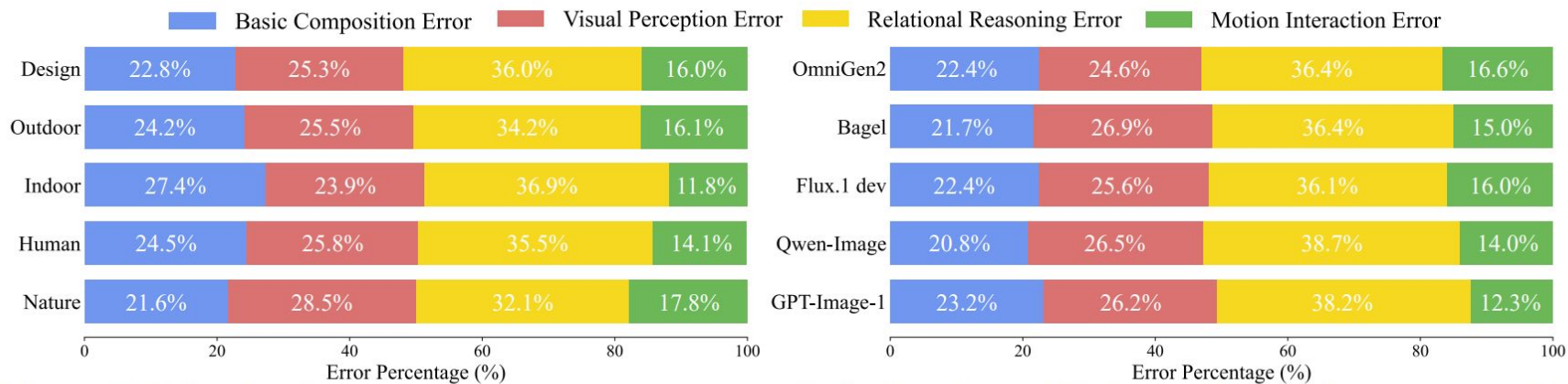


Figure 5: Distribution of error types across scenes (left, based on all T2I models) and some examples of T2I models (right) in our SpatialGenEval.



04

THEIR SOLUTION

THE SPATIALT2I Dataset

Why build it?

The benchmark diagnoses the problem. SpatialT2I asks: can the same information-dense design philosophy fix it?

How it was built

- **Stage 1: Prompt and Omni-dimensional QAs generation:** Each of 1,100 filtered prompts was given to 14 top T2I models (those scoring above 50% on spatial QA) -- yielding 15,400 image-text pairs
- **Stage 2: Rewrite prompt to obtain text-image pair** -- prompt rewriting: Instead of discarding mismatched images, Gemini 2.5 Pro rewrites the prompt to match what the image actually shows -- preserving spatial density
- All 15,400 pairs annotated across 10 spatial sub-domains via 10 MCQ entries each. Human experts reviewed for logical consistency.

SFT Results

Training setup

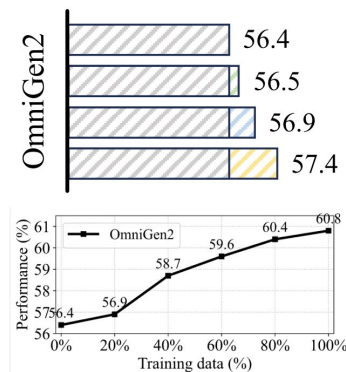
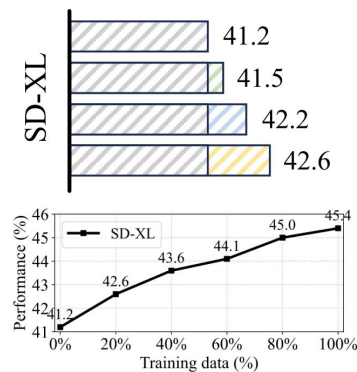
- Models: Stable Diffusion-XL, UniWorld-V1, OmniGen2
- AdamW optimizer, lr 5e-6, batch size 16, 3 epochs, 8x A100 GPUs. Loss = image reconstruction + QA cross-entropy (weighted 1.0 + 0.5)

Results

- SD-XL: 41.2% → 45.4% (+4.2%)
- UniWorld-V1: 54.2% → 59.9% (+5.7%)
- OmniGen2: 56.4% → 60.8% (+4.4%)

Where the gains come from

- Biggest gains in: position, layout, comparison
- Smaller gains in: occlusion and causality -- these bottlenecks persist even after fine-tuning
- **Takeaway:** No architecture redesign needed -- better spatial training data alone yields consistent gains across all model families



Prompt 1: In a forest, a **grey squirrel** on a **left branch** of a massive oak tree **drops an acorn** towards a red squirrel at the bottom-center. The two squirrels **face each other**. Above them, a **wise-looking owl** with yellow eyes, **partially obscured** by its tree hole, faces the observer. The oak tree is much **taller than** a nearby mossy log. The red squirrel is **close to** a **small pile of three brown acorns**, illuminated by a sunbeam.



GPT-Image-1



Qwen-Image



Flux.1 dev



Bagel



UniWorld-V1



OmniGen2



+SpatialT2I



Prompt 2: In a forest clearing, a family of **three—two adults** with **a child between them**—sits on a **red and white checkered blanket**. They **face away** from the viewer, looking up at fireworks. A golden retriever lies **next to** the child, and a picnic basket is on the **blanket's right side**. A tall tree on the left **partially blocks** the fireworks. One **adult pulls the child close** as a gentle **wind rustles the leaves**.

05

CONCLUSION

Key Takeaways

Three clean takeaways:

1. **The ceiling is 60%.** Even the best models get 4 in 10 spatial questions wrong. This is a fundamental capability gap, not an edge case.
2. **The bottleneck is reasoning, not perception.** Models handle "what is in the scene" reasonably well. They collapse on "where exactly," "how are they related," and "why does this interaction happen" — especially occlusion, comparison, and causality.
3. **Data-centric fine-tuning works — and is accessible.** You don't need a new architecture. Fine-tuning on 15,400 information-dense pairs with a QA-guided loss yields consistent +4–6% gains. The recipe is repeatable and open-source.

06

THANK YOU!